Devesh Gautam

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	8.63
Sarvodaya Senior Secondary School, Kota	Class XII (CBSE)	2023	96.00%
Sarvodaya Senior Secondary School, Kota	Class X (CBSE)	2021	88.60%

SCHOLASTIC ACHIEVEMENTS

- **Expert** on codeforces with 1608 rating
- **NTSE Stage II Scholar:** Selected among top 2,000 students nationwide in the prestigious National Talent Search Examination (NTSE) conducted by NCERT, India

EXPERIENCE

Sprinklr | ML Engineering Intern

May 2026 - July 2026

- Architected an automated **Golden Test Set (GTS)** generation framework to evaluate custom RAG systems, scaling robust evaluation pipelines across 40 distinct client datasets while significantly reducing latency compared to open-source baselines like RAGAS.
- Engineered a **semantic clustering pipeline** to synthesize **8 distinct QA categories**--including inter/intra-document multi-hop, hard negatives, and comparative queries--by selectively pairing semantically related chunks from isolated vector spaces.
- Formulated a granular **root-cause analysis (RCA)** methodology that decomposes ground-truth answers into atomic facts, systematically **isolating pipeline failure** modes across chunking, retrieval, reranking, and generation stages.
- **Benchmarked** production RAG deployments utilizing hybrid retrieval (BM25 and dense similarity search) and cross-encoding, evaluating model precision, recall, and infrastructure costs against the generated GTS.

PROJECTS

Image Compression using K-Means, Mini-Batch K-Means, and GMM | Python, scikit-learn, matplotlib

- Implemented unsupervised clustering techniques (K-Means, Mini-Batch K-Means, GMM) to compress RGB images by reducing the color palette based on cluster centers
- Highlighted performance differences across hard (K-Means) and soft (GMM) clustering methods using statistical and visual analysis in Python

Probabilistic NLP Framework for Airline Review Analysis

- Extracted quantifiable ratings and insights from airline review using a custom probabilistic NLP model.
- Built a probabilistic model to derive confidence metrics suggest optimal, data-driven travel itineraries.
- Provided data-driven insights and recommendations by analyzing thousands of customer feedback entries.

C++ Adaptive Cache Replacement (ARC) Simulator | Core Data Structures & Algorithms Course Project

- Designed and implemented multiple cache policies (LRU, LFU, MRU, ARC) in C++ with an object-oriented modular architecture
- Benchmarked all policies using real-world disk I/O traces (MSR Cambridge) across cache sizes (100-5000 entries)
- Analyzed hit ratio and latency metrics to evaluate performance

Applied Stats course project - Quantifying Inequity Through Comparative Analysis

- Performed comparative statistical analysis on Nashville policing data using two-sample inference methods, revealing significant racial disparities in traffic stop rates after adjusting for population size and variance.

- Applied statistical modeling techniques--including Gamma distribution fitting, Bernoulli hypothesis testing, confidence intervals, chi-squared, and t-distribution analyses--on real and synthetic datasets to extract robust population-level insights.

SKILLS

- **Programming Languages:** Proficiency in C, C++ & Python. Familiar with SQL
- **Software & Tools:** Jupyter Notebook, LaTeX, GitHub, VS Code
- **Frameworks & Libraries:** Numpy, Pandas, Matplotlib, Keras, Tensorflow, SciPy, OpenCV, NLTK, SpaCy

RELEVANT COURSES

- **Mathematics :** Data Structures and Applications Lab, Number Theory, Probability Theory, Applied Statistics, Combinatorics and Graph Theory, Calculus, Linear Algebra, Introduction to Metric Spaces, Introduction to Group Theory, Ordinary Differential Equations
- **Computer Science:** Data Structures and Applications, Algorithms, Database Management Systems, Compilers, Introduction to Programming, Theory of Computation
- **Miscellaneous:** Introduction to Entrepreneurship, Communication Skills

EXTRACURRICULAR

- **Co-authored** a **mathematics book** as part of Writing and Presenting Mathematics course; presented chapters to faculty
- **Team member** of department **cricket team** in **Diesta** (Inter-Departmental Championship)

POSITIONS OF RESPONSIBILITY

- **Core,** Infero- Competitive coding club
- **Deputy Sponsorship Coordinator,** Elan & Envision - The techno cultural fest